

Intermediate Elective 2

Minh Tri Ngo

Set Up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)

# imported libraries from example
library(showtext)
library(camcorder)
library(ggtext)
library(glue)
library(ggfx)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
  ↪ Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
```

```

group_by(date_site) |>
# calculate median pH, salinity, DO
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
             names_to = "taxon",
             values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
  ↪ liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>

```

```

# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
  ↪   = TRUE),
  site_full = fct_rev(site_full))

# creating new object from clean aquatic inverts
copepods_df <- aq_ins_clean |>
# filter to only include copepods
filter(taxon == "copepod") |>
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))

# creating new clean object from aquatic inverts
ab_do <- aq_ins_clean |>
# log transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit+1),
  # convert taxon to title
  taxon = to_any_case(taxon, case = "title"),
  # order taxon levels by abundance
  taxon = fct_reorder(taxon, ave_lit, .fun="max"))

```

1. Explain Inspiration

- The visualization I chose for my inspiration was the “Can an exploding snowman predict the summer season” by Nicola Rennie.
- I felt that this visualization makes sense as I would like to analyze temperature in relation to the different taxon availability and the graph allows me to analyze and communicate my findings the most efficiently.
- The variables that I will choose in my dataset to show my visualization will be dissolved, taxon, and average abundance. In terms of visual components, the dissolved will be on the x axis. Taxon will have different colors to differentiate. And the y axis will have the average abundance.

2. Plan figure

3. Code figure

```
# Load fonts -----  
  
font_add_google("Poppins", "Poppins")  
showtext_auto()  
showtext_opts(dpi = 300)  
title_font <- "Poppins"  
body_font <- "Poppins"  
  
# Define colours and fonts-----  
  
bg_col <- "#F2F4F8"  
text_col <- "#151C28"  
highlight_col <- "cornflowerblue"  
  
# Taxon color palette (one color per taxon)  
taxon_colors <- c(  
  "Annelida Oligochaete"      = "#2D6A4F",  
  "Annelida Polychaete"      = "#52B788",  
  "Cladocera"                 = "#74C69D",  
  "Copepod"                   = "#B7E4C7",  
  "Diptera Ceratopogonidae"   = "#D4A373",  
  "Diptera Chironomid"       = "#A0522D",
```

```

"Hemiptera Corixidae Boatman" = "#6A3D9A",
"Nematode"                    = "#1D78B4",
"Ostracod"                    = "#E63946"
)

# Data prep -----

plot_data <- ab_do |>
  filter(!is.na(med_do), !is.na(log_ave_lit))

med_do_val <- median(plot_data$med_do,      na.rm = TRUE)
med_ab_val <- 3

# Start recording -----

gg_record(
  dir = file.path("2025", "2025-12-02", "recording"),
  device = "png",
  width = 7,
  height = 6,
  units = "in",
  dpi = 300
)

# Define text -----

social <- "Your Name"
icon <- "<span
  ↪ style='font-size:8pt;font-family:\"FA\";color:#3C598E;'>&#xf111;</span> "
title <- "Dissolved Oxygen and Aquatic Invertebrate Abundance"
st <- glue("Using a log-transformed average abundance per liter plotted
  ↪ against median dissolved oxygen (mg/L) across NCOS quarterly sites.")
cap <- paste0(
  "**Data**:", NCOS Aquatic Sampling 2021-2026 | **Graphic: Minh Tri Ngo",
  ↪ social
)

most_abundant <- ab_do |>
  slice_max(log_ave_lit)

```

```

highest_do <- ab_do |>
  slice_max(med_do)

# Plot -----

ggplot(
  data = ab_do          # <-- swapped: was `plot_data`
) +
  annotate(
    "rect",
    xmin = median(ab_do$med_do), xmax = Inf,          # <-- swapped: was
    ↪ `plot_data$duration`
    ymin = median(ab_do$log_ave_lit), ymax = Inf, alpha = 0.5, # <--
    ↪ swapped: was `plot_data$tre200m0`
    fill = "grey"
  ) +
  annotate(
    "text",
    x = 15, y = 5.5, label = "High D0, High abundance",          # <--
    ↪ swapped: coordinates + label
    colour = "grey30", family = body_font, fontface = "bold.italic",
    hjust = 1.05
  ) +
  annotate(
    "rect",
    xmin = median(ab_do$med_do), xmax = Inf,          # <-- swapped
    ymin = -Inf, ymax = median(ab_do$log_ave_lit), alpha = 0.5, # <--
    ↪ swapped
    fill = highlight_col
  ) +
  annotate(
    "text",
    x = 15, y = 0.5, label = "High D0, Low abundance",          # <--
    ↪ swapped
    colour = "#3C598E", family = body_font, fontface = "bold.italic",
    hjust = 1.05
  ) +
  annotate(
    "rect",
    xmin = -Inf, xmax = median(ab_do$med_do),          # <-- swapped
    ymin = median(ab_do$log_ave_lit), ymax = Inf, alpha = 0.5, # <--
    ↪ swapped
  )

```

```

    fill = highlight_col
  ) +
  annotate(
    "text",
    x = 0.5, y = 5.5, label = "Low D0,\nHigh\nabundance",      # <--
    ↪ swapped
    colour = "#3C598E", family = body_font, fontface = "bold.italic",
    hjust = 0
  ) +
  annotate(
    "rect",
    xmin = -Inf, xmax = median(ab_do$med_do),      # <-- swapped
    ymin = -Inf, ymax = median(ab_do$log_ave_lit), alpha = 0.5, # <--
    ↪ swapped
    fill = "grey"
  ) +
  annotate(
    "text",
    x = 0.5, y = 0.5, label = "Low D0,\nLow\nabundance",      # <--
    ↪ swapped
    colour = "grey30", family = body_font, fontface = "bold.italic",
    hjust = 0
  ) +
  with_outer_glow(
    geom_point(
      mapping = aes(
        x = med_do, y = log_ave_lit,      # <-- swapped: was `duration`,
        ↪ `tre200m0`
        colour = taxon                    # <-- swapped: was `correct`; removed
        ↪ `shape`
      ),
      size = 3
    ),
    colour = "white",
    expand = 3
  ) +
  scale_colour_manual(values = c(      # <-- swapped: your 9 taxon colors
    "Annelida Oligochaete" = "#2D6A4F",
    "Annelida Polychaete" = "#52B788",
    "Cladocera" = "#74C69D",
    "Copepod" = "#B7E4C7",
    "Diptera Ceratopogonidae" = "#D4A373",

```

```

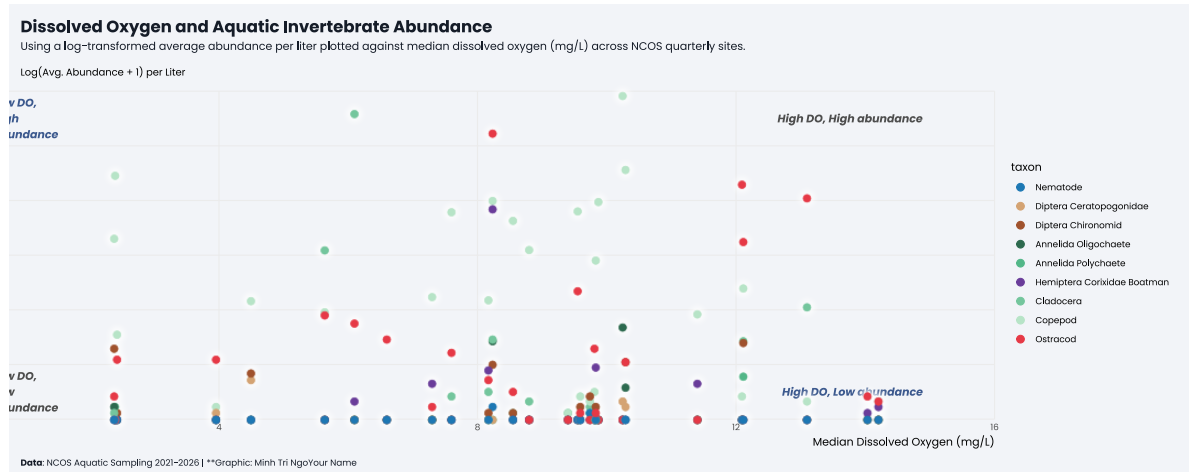
    "Diptera Chironomid"          = "#A0522D",
    "Hemiptera Corixidae Boatman" = "#6A3D9A",
    "Nematode"                    = "#1D78B4",
    "Ostracod"                    = "#E63946"
  )) +
  scale_x_continuous(limits = c(0, 16)) +      # <-- swapped: D0 range ~0-15
  ↪ mg/L
  scale_y_continuous(limits = c(0, 6)) +      # <-- swapped: log(ave_lit+1)
  ↪ range
  labs(
    x = "Median Dissolved Oxygen (mg/L)",      # <-- swapped
    y = "Log(Avg. Abundance + 1) per Liter",   # <-- swapped
    title = title, subtitle = st,
    caption = cap
  ) +
  coord_cartesian(expand = FALSE, clip = "off") +
  theme_minimal(base_size = 10, base_family = body_font) +
  theme(
    legend.position = "right",                 # <-- swapped: "none" → "right"
    ↪ to show taxon legend
    plot.margin = margin(10, 20, 10, 10),
    plot.title.position = "plot",
    plot.caption.position = "plot",
    plot.background = element_rect(fill = bg_col, colour = bg_col),
    panel.background = element_rect(fill = bg_col, colour = bg_col),
    plot.title = element_textbox_simple(
      colour = text_col,
      hjust = 0,
      halign = 0,
      margin = margin(b = 0, t = 5),
      family = title_font,
      face = "bold",
      size = rel(1.5)
    ),
    plot.subtitle = element_textbox_simple(
      colour = text_col,
      hjust = 0,
      halign = 0,
      margin = margin(b = 35, t = 5),
      family = body_font
    ),
    plot.caption = element_textbox_simple(
      colour = text_col,

```

```

    hjust = 0,
    halign = 0,
    margin = margin(b = 0, t = 10),
    family = body_font
  ),
  strip.text = element_textbox_simple(
    face = "bold",
    margin = margin(t = 10),
    size = rel(0.9)
  ),
  panel.grid.minor = element_blank(),
  axis.title.x = element_text(hjust = 1, margin = margin(t = 5)),
  axis.title.y.left = element_text(
    angle = 0, vjust = 1.07,
    family = body_font,
    size = rel(0.9),
    margin = margin(r = -215)
  )
)
)

```



```

ggplot(
  data = plot_data
) +
  annotate(
    "rect",
    xmin = med_do_val, xmax = Inf,
    ymin = med_ab_val, ymax = Inf, alpha = 0.5,
    fill = "#74C69D"
  )

```

```

) +
annotate(
  "text",
  x = 15.5, y = 5.8, label = "High D0, High abundance",
  colour = "#2D6A4F", family = body_font, fontface = "bold.italic",
  hjust = 1.05
) +
annotate(
  "rect",
  xmin = med_do_val, xmax = Inf,
  ymin = -Inf, ymax = med_ab_val, alpha = 0.5,
  fill = highlight_col
) +
annotate(
  "text",
  x = 15.5, y = 0.2, label = "High D0, Low abundance",
  colour = "#3C598E", family = body_font, fontface = "bold.italic",
  hjust = 1.05
) +
annotate(
  "rect",
  xmin = -Inf, xmax = med_do_val,
  ymin = med_ab_val, ymax = Inf, alpha = 0.5,
  fill = "#D4A373"
) +
annotate(
  "text",
  x = 0.3, y = 5.8, label = "Low D0,\nHigh\nabundance",
  colour = "#A0522D", family = body_font, fontface = "bold.italic",
  hjust = 0
) +
annotate(
  "rect",
  xmin = -Inf, xmax = med_do_val,
  ymin = -Inf, ymax = med_ab_val, alpha = 0.5,
  fill = "grey"
) +
annotate(
  "text",
  x = 0.3, y = 0.2, label = "Low D0,\nLow\nabundance",
  colour = "grey30", family = body_font, fontface = "bold.italic",
  hjust = 0
) +

```

```

with_outer_glow(
  geom_point(
    mapping = aes(
      x = med_do, y = log_ave_lit,
      colour = taxon
    ),
    size = 3
  ),
  colour = "white",
  expand = 3
) +
scale_colour_manual(values = c(
  "Annelida Oligochaete"      = "#2D6A4F",
  "Annelida Polychaete"      = "#52B788",
  "Cladocera"                 = "#74C69D",
  "Copepod"                   = "#B7E4C7",
  "Diptera Ceratopogonidae"   = "#D4A373",
  "Diptera Chironomid"        = "#A0522D",
  "Hemiptera Corixidae Boatman" = "#6A3D9A",
  "Nematode"                  = "#1D78B4",
  "Ostracod"                  = "#E63946"
)) +
scale_x_continuous(limits = c(0, 16)) +
scale_y_continuous(limits = c(0, 6)) +
labs(
  x = "Median Dissolved Oxygen (mg/L)",
  y = "Log(Avg. Abundance + 1)\nper Liter",
  title = title, subtitle = st,
  caption = cap
) +
coord_cartesian(expand = FALSE, clip = "off") +
theme_minimal(base_size = 10, base_family = body_font) +
theme(
  legend.position = "right",
  plot.margin = margin(10, 20, 10, 50),
  plot.title.position = "plot",
  plot.caption.position = "plot",
  plot.background = element_rect(fill = bg_col, colour = bg_col),
  panel.background = element_rect(fill = bg_col, colour = bg_col),
  plot.title = element_textbox_simple(
    colour = text_col,
    hjust = 0,
    halign = 0,

```

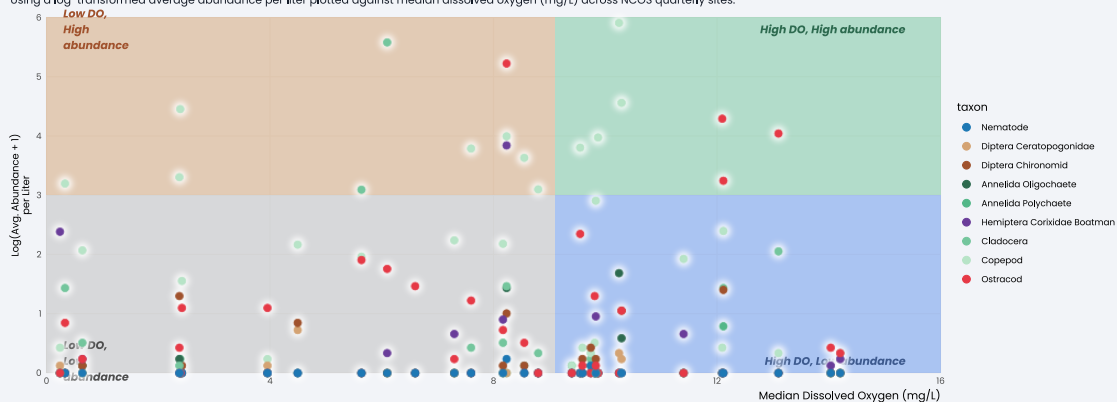
```

    margin = margin(b = 0, t = 5),
    family = title_font,
    face = "bold",
    size = rel(1.5)
),
plot.subtitle = element_textbox_simple(
  colour = text_col,
  hjust = 0,
  halign = 0,
  margin = margin(b = 10, t = 5),
  family = body_font
),
plot.caption = element_textbox_simple(
  colour = text_col,
  hjust = 0,
  halign = 0,
  margin = margin(b = 0, t = 10),
  family = body_font
),
strip.text = element_textbox_simple(
  face = "bold",
  margin = margin(t = 10),
  size = rel(0.9)
),
panel.grid.minor = element_blank(),
axis.title.x = element_text(hjust = 1, margin = margin(t = 5)),
axis.title.y.left = element_text(
  angle = 90,
  vjust = 0.5,
  family = body_font,
  size = rel(0.9),
  margin = margin(r = 5)
)
)

```

Dissolved Oxygen and Aquatic Invertebrate Abundance

Using a log-transformed average abundance per liter plotted against median dissolved oxygen (mg/L) across NCOS quarterly sites.



Data: NCOS Aquatic Sampling 2021-2026 | **Graphic: Minh Tri NgoYour Name